A pulse-based diet is effective for reducing total and LDL-cholesterol in older adults.

Our purpose was to determine the effects of a pulse-based diet in individuals 50 years or older for reducing CVD risk factors. A total of 108 participants were randomised to receive pulse-based foods (two servings daily of beans, chickpeas, peas or lentils; about 150 g/d dry weight) or their regular diet for 2 months, followed by a washout of 1 month and a cross-over to the other diet for 2 months. Anthropometric measures, body composition and biochemical markers (i.e. serum LDL-cholesterol (LDL-C), as the primary outcome, and other lipids, glucose, insulin and C-reactive protein) were assessed before and after each diet phase. A total of eighty-seven participants (thirty males and fifty-seven females; 59·7 (sd 6·3) years, body mass 76 (sd 16) kg) completed the study. Compared with the regular diet, the pulse-based diet decreased total cholesterol by 8·3 % (pulse, 4·57 (sd 0·93) to 4·11 (sd 0·91) mmol/l; regular, 4·47 (sd 0·94) to 4·39 (sd 0·97) mmol/l; $P < 0·001$) and LDL-C by 7·9 % (pulse, 2·93 (sd 0·84) to 2·55 (sd 0·75) mmol/l; regular, 2·96 (sd 0·86) to 2·81 (sd 0·83) mmol/l; $P = 0·01$). In a sub-analysis of individuals with high lipid levels at baseline (twenty individuals with high cholesterol), the pulse-based diet reduced cholesterol by 6 % compared with the regular diet (pulse, 5·62 (sd 0·78) to 5·26 (sd 0·68) mmol/l; regular, 5·60 (sd 0·91) to 5·57 (sd 0·85) mmol/l; $P = 0·05$). A pulse-based diet is effective for reducing total cholesterol and LDL-C in older adults and therefore reduces the risk of CVD.